



**F-105**

**M. B. A. (Second Semester) Examination, May 2026**

**(New Course)**

**BUSINESS RESEARCH METHODS & DECISION  
SCIENCE**

***Paper : FT-2025205***

***Time Allowed : Three hours***

***Maximum Marks : 85***

***Note : Attempt any five questions. All questions  
carry equal marks. Draw neat diagrams,  
wherever necessary. Table values are provided  
and already given in the relevant numericals.***

1. Define the term Research. Explain any three types of Social Science Research. As an MBA student which one you like the most and why? Give 2 reasons?



2. What do you mean by Research Design? Explain briefly all steps involved in the Research process.
3. Describe the various sources of Primary and Secondary Data. Which one you like the most and why?
4. Discuss any **two** :
  - (i) Scaling techniques
  - (ii) ANOVA test
  - (iii) Observation method
  - (iv) Questionnaires
5. What do you mean by Game Theory? Explain in brief the conceptual framework of Game Theory.
6. Write a detailed note on – “Research Report Writing”. Narrate the main components of a Research Report.
7. What is Decision Science? Explain role and importance of Decision Science in decision making.
8. Calculate Karl Pearson’s correlation co-efficient between “Length” and “Weight” from the given data and interpret



about "High"/or "Low" dedgree of positive correlation between length and weight :

|                 |    |     |     |     |    |
|-----------------|----|-----|-----|-----|----|
| Length (inches) | 3, | 4,  | 6,  | 7,  | 10 |
| Weight (ounces) | 9, | 11, | 14, | 15, | 16 |

(Assuming - "X" denoting "Length" and "Y" - "Weight")

9. A sample of 10 students has an average mean score of 52 marks. The population mean is 50 marks and the sample standard deviation is 4 marks. (Carefully note - that as per "*t*-table" at  $\alpha = 0.05$  for 9 degrees of freedom  $df = 9$ , the critical value as per table  $df = \pm 2.262$ ). Test at 5% Level of significance, whether the sample mean is significantly different from the population mean from above information and given critical value of  $t = \text{table} = \pm 2.262$ . You are required to calculate *t*-value.

(Read carefully the given *t*-table value  $\pm 2.262$

$\rightarrow \alpha = 0.05 \rightarrow \text{degree freeom } df = 9$ )

10. (a) Calculate "Median" and "Mode" from the following series mean :

$8\frac{1}{2}$



| Marks | No. of Students |
|-------|-----------------|
| 0-10  | 15              |
| 10-20 | 25              |
| 20-30 | 32              |
| 30-40 | 56              |
| 40-50 | 78              |
| 50-60 | 80              |
| 60-70 | 70              |

(b) A die is thrown 132 times with following result :  $8\frac{1}{2}$

|          |              |    |    |    |    |    |    |
|----------|--------------|----|----|----|----|----|----|
| Result : | No. Turnedup | 1  | 2  | 3  | 4  | 5  | 6  |
|          | Frequency    | 16 | 20 | 25 | 14 | 29 | 28 |

Test the hypothesis that die is unbiased using chi-square ( $\chi^2$  test) for 5 degrees of freedom at 5% level = the value of chi-square table is given

$$\chi^2 = 11.070.$$

(Carefully Read the  $df = 5$ ,  $\chi^2$  table value given

$$\chi^2 = 11.070)$$

